

Question 1 _MODULE 5 Assignment

rohit45@rohit45-VirtualBox:~\$ cd dir1

rohit45@rohit45-VirtualBox:~/dir1\$ ls

dir2 f1.txt f2.txt f3.sh f4.csv f.txt smlinkfile.txt

rohit45@rohit45-VirtualBox:~/dir1\$ cat smlinkfile.txt

Original File @ /dir2

rohit45@rohit45-VirtualBox:~/dir1\$ cd dir2

rohit45@rohit45-VirtualBox:~/dir1/dir2\$ ls

orgfile.txt

rohit45@rohit45-VirtualBox:~/dir1/dir2\$ cat orgfile.txt

Original File @ /dir2

rohit45@rohit45-VirtualBox:~/dir1/dir2\$ cd ~

rohit45@rohit45-VirtualBox:~\$ cat file_analyzer.sh

#!/bin/bash

FOUND_GLOBAL=0

rec_filesearch()

```
{  
    dir="$1"  
    keyword="$2"  
    echo "PROCESSING RECURSIVE SEARCH TO FIND $keyword in $dir"  
    for i in "$dir"/*;  
    do  
        # echo "PROCESSING RECURSIVE SEARCH TO FIND $keyword in $dir"  
        if [[ -f "$i" ]];then  
            # echo "IN FILE $i"  
            grep -E "$keyword" "$i"
```

```

if [[ "$?" -eq 0 ]];then
#      echo "-----"
      FOUND_GLOBAL=1
      echo "KEYWORD FINDING SUCCESS"
      echo "PATH : $i"
      echo "-----"
      echo

fi

elif [[ -d "$i" ]];then
#      echo "IN SUBDIR $i"
      rec_filesearch "$i" "$keyword"

elif [[ -s "$i" ]];then
      echo "ENTERED AN EMPTY FILE:$i"

fi

done
}

```

```

logger(){
      echo -e "$(date)\n$1\n" | tee -a errors.log >&2
}

```

here_doc()

```
{
```

```
      cat <<EOF
```

MENU USING HERE DOCUMENT

```
-----
```

In Script : \$0

```
-----
```

Operations allowed :

-d <dir_name> for directory
-f <file_name> for file
-k <keyword> for Keyword Search
--help Display Help Menu

EOF

}

here_str()

{

echo "-----"

echo "IN HERE_STR FN WITH INPUT FILE "\$1" AND KEYWORD "\$2""

if [[! -f "\$1"]];then

logger "File not found in here_str:\$1"

exit 1

else

grep -E "\$2" <<<"\$(cat "\$1")"

if [["\$?" -ne 0]];then

**logger "KEYWORD \$2 IS NOT AVAILABLE IN \$1 (FROM
HERE_STR)"**

fi

fi

echo "-----"

}

validate_ipkeyword()

{

if [[! "\$1" =~ ^[[:alnum:]]+\$]];then

logger "Improper Keyword Entry : "\$1""

exit 1

fi

```
}
```

```
if [[ "$1" == "--help" ]];then
```

```
    here_doc
```

```
    exit 0
```

```
fi
```

```
while getopts ":d:f:k:" opt;
```

```
do
```

```
    case "$opt" in
```

```
        d)direc=$OPTARG;;
```

```
        f)file=$OPTARG;;
```

```
        k)keyw=$OPTARG;;
```

```
        ?)logger "INVALID OPTION ENTERED IN GETOPTS $OPTARG"; exit 1;;
```

```
        :)logger "NO ARG FOR GETOPTS $OPTARG"; exit 1;;
```

```
    esac
```

```
done
```

```
echo "IN SCRIPT: $0"
```

```
echo "TOTAL ARGS : $#"
```

```
echo "ARGS ENTERED : $@"
```

```
echo
```

```
if [[ -z "$keyw" ]];then
```

```
    logger "KEYWORD NOT PROVIDED"
```

```
    exit 1
```

```
fi
```

```
validate_ipkeyword "$keyw"
```

```

if [[ -n "$direc" ]];then
    rec_filesearch "$direc" "$keyw"
    if [[ "$FOUND_GLOBAL" -eq 0 ]]; then
        logger "Keyword $keyw not found in Recursive search"
    fi
elif [[ -n "$fle" ]];then
    here_str "$fle" "$keyw"
else
    logger "NO VALID INPUT PROVIDED"
    here_doc
    exit 1
fi

```

```
#validate_ipkeyword "@#123"
```

```
#rec_filesearch "dir1" "File"
```

```
#here_str "dir1/dir2/orgfile.txt" "File"
```

```
rohit45@rohit45-VirtualBox:~$ ./file_analyzer.sh --help
```

MENU USING HERE DOCUMENT

In Script : ./file_analyzer.sh

Operations allowed :

-d <dir_name> for directory

-f <file_name> for file

-k <keyword> for Keyword Search

--help Display Help Menu

```
rohit45@rohit45-VirtualBox:~$ ./file_analyzer.sh -d dir1 -k File
```

IN SCRIPT: ./file_analyzer.sh

TOTAL ARGS : 4

ARGS ENTERED : -d dir1 -k File

PROCESSING RECURSIVE SEARCH TO FIND File in dir1

PROCESSING RECURSIVE SEARCH TO FIND File in dir1/dir2

Original File @ /dir2

KEYWORD FINDING SUCCESS

PATH : dir1/dir2/orgfile.txt

Original File @ /dir2

KEYWORD FINDING SUCCESS

PATH : dir1/slinkfile.txt

rohit45@rohit45-VirtualBox:~\$./file_analyzer.sh -d dir1 -k MEPCO

IN SCRIPT: ./file_analyzer.sh

TOTAL ARGS : 4

ARGS ENTERED : -d dir1 -k MEPCO

PROCESSING RECURSIVE SEARCH TO FIND MEPCO in dir1

PROCESSING RECURSIVE SEARCH TO FIND MEPCO in dir1/dir2

Thu Feb 5 11:06:26 PM IST 2026

Keyword MEPCO not found in Recursive search

rohit45@rohit45-VirtualBox:~\$ cat errors.log

Thu Feb 5 11:06:26 PM IST 2026

Keyword MEPCO not found in Recursive search

rohit45@rohit45-VirtualBox:~\$./file_analyzer.sh -f dir1/dir2/orgfile.txt -k File

IN SCRIPT: ./file_analyzer.sh

TOTAL ARGS : 4

ARGS ENTERED : -f dir1/dir2/orgfile.txt -k File

IN HERE_STR FN WITH INPUT FILE dir1/dir2/orgfile.txt AND KEYWORD File

Original File @ /dir2

rohit45@rohit45-VirtualBox:~\$./file_analyzer.sh -f dir1/dir2/orgfile.txt -k MEPCO

IN SCRIPT: ./file_analyzer.sh

TOTAL ARGS : 4

ARGS ENTERED : -f dir1/dir2/orgfile.txt -k MEPCO

IN HERE_STR FN WITH INPUT FILE dir1/dir2/orgfile.txt AND KEYWORD MEPCO

Thu Feb 5 11:07:15 PM IST 2026

KEYWORD MEPCO IS NOT AVAILABLE IN dir1/dir2/orgfile.txt (FROM HERE_STR)

rohit45@rohit45-VirtualBox:~\$ cat errors.log

Thu Feb 5 11:06:26 PM IST 2026

Keyword MEPCO not found in Recursive search

Thu Feb 5 11:07:15 PM IST 2026

KEYWORD MEPCO IS NOT AVAILABLE IN dir1/dir2/orgfile.txt (FROM HERE_STR)

rohit45@rohit45-VirtualBox:~\$./file_analyzer.sh -d dir1 -k @#123

IN SCRIPT: ./file_analyzer.sh

TOTAL ARGS : 4

ARGS ENTERED : -d dir1 -k @#123

Thu Feb 5 11:07:40 PM IST 2026

Improper Keyword Entry : @#123

rohit45@rohit45-VirtualBox:~\$ cat errors.log

Thu Feb 5 11:06:26 PM IST 2026

Keyword MEPCO not found in Recursive search

Thu Feb 5 11:07:15 PM IST 2026

KEYWORD MEPCO IS NOT AVAILABLE IN dir1/dir2/orgfile.txt (FROM HERE_STR)

Thu Feb 5 11:07:40 PM IST 2026

Improper Keyword Entry : @#123

rohit45@rohit45-VirtualBox:~\$./file_analyzer.sh -f wrong.txt -k File

IN SCRIPT: ./file_analyzer.sh

TOTAL ARGS : 4

ARGS ENTERED : -f wrong.txt -k File

IN HERE_STR FN WITH INPUT FILE wrong.txt AND KEYWORD File

Thu Feb 5 11:07:54 PM IST 2026

File not found in here_str:wrong.txt

rohit45@rohit45-VirtualBox:~\$ cat errors.log

Thu Feb 5 11:06:26 PM IST 2026

Keyword MEPCO not found in Recursive search

Thu Feb 5 11:07:15 PM IST 2026

KEYWORD MEPCO IS NOT AVAILABLE IN dir1/dir2/orgfile.txt (FROM HERE_STR)

Thu Feb 5 11:07:40 PM IST 2026

Improper Keyword Entry : @#123

Thu Feb 5 11:07:54 PM IST 2026

File not found in here_str:wrong.txt

rohit45@rohit45-VirtualBox:~\$./file_analyzer.sh -d dir1

SCRIPT: ./file_analyzer.sh

TOTAL ARGS : 2

ARGS ENTERED : -d dir1

Thu Feb 5 11:08:30 PM IST 2026

KEYWORD NOT PROVIDED

rohit45@rohit45-VirtualBox:~\$ cat errors.log

Thu Feb 5 11:06:26 PM IST 2026

Keyword MEPCO not found in Recursive search

Thu Feb 5 11:07:15 PM IST 2026

KEYWORD MEPCO IS NOT AVAILABLE IN dir1/dir2/orgfile.txt (FROM HERE_STR)

Thu Feb 5 11:07:40 PM IST 2026

Improper Keyword Entry : @#123

Thu Feb 5 11:07:54 PM IST 2026

File not found in here_str:wrong.txt

Thu Feb 5 11:08:30 PM IST 2026

KEYWORD NOT PROVIDED

rohit45@rohit45-VirtualBox:~\$./file_analyzer.sh

SCRIPT: ./file_analyzer.sh

TOTAL ARGS : 0

ARGS ENTERED :

Thu Feb 5 11:08:43 PM IST 2026

KEYWORD NOT PROVIDED

rohit45@rohit45-VirtualBox:~\$ cat errors.log

Thu Feb 5 11:06:26 PM IST 2026

Keyword MEPCO not found in Recursive search

Thu Feb 5 11:07:15 PM IST 2026

KEYWORD MEPCO IS NOT AVAILABLE IN dir1/dir2/orgfile.txt (FROM HERE_STR)

Thu Feb 5 11:07:40 PM IST 2026

Improper Keyword Entry : @#123

Thu Feb 5 11:07:54 PM IST 2026

File not found in here_str:wrong.txt

Thu Feb 5 11:08:30 PM IST 2026

KEYWORD NOT PROVIDED

Thu Feb 5 11:08:43 PM IST 2026

KEYWORD NOT PROVIDED